

$$\begin{array}{ccc}
(1, a_1) & \xrightarrow{\psi_0} & F^T 1 \\
j_{F^T X}^T \downarrow & & \downarrow F^T j_X \\
[F^T X, F^T X]_T & \xleftarrow{\psi_{X, X}} & F^T [X, X]
\end{array}$$